School of Engineering & Technology

Department of Computer Science & Technology

Agentic AI Lab (CSCR3214)

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for

B. Tech. (CSE)

6th Semester

Lab Faculty details:

Ayush Kumar Singh

Department of Computer Science & Engineering
School of Engineering & Technology Sharda University

Student details:

Taushir Alam

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I. INTRODUCTION

Artificial Intelligence (AI) has emerged as a transformative force in the healthcare sector, fundamentally altering how diseases are diagnosed, treated, and managed. The integration of machine learning algorithms, natural language processing, and computer vision into clinical workflows aims to enhance the precision of medical interventions and improve patient outcomes. Historically, healthcare relied on manual data entry and reactive treatment plans; however, the shift toward proactive, data-driven medicine is now powered by AI’s ability to analyze massive datasets beyond human capability. The purpose of this report is to evaluate how AI is currently being utilized in clinical settings, the efficiencies it brings to hospital administration, and the profound impact it has on reducing human error in high-stakes medical environments.

II. KEY FINDINGS

- **Diagnostic Accuracy:** AI algorithms for medical imaging (Radiology/Pathology) have reached a **90-95% accuracy rate** in identifying early-stage tumors, often detecting anomalies 18-24 months before human observation.

- **Drug Discovery:** AI-driven platforms have reduced the time required for the "target identification" phase of drug development from **5 years to approximately 12 months**.

- **Predictive Analytics:** Hospitals using AI-driven patient monitoring have seen a **25% reduction in unplanned ICU admissions** by predicting sepsis and cardiac arrest hours in advance.

- **Administrative Efficiency:** NLP-based voice-to-text tools have reduced clinical documentation time for doctors by **3 hours per day**, significantly lowering "physician burnout."

III. CHALLENGES

The primary challenge facing AI in healthcare is **Data Privacy and Security**. Healthcare data is highly sensitive, and the use of large language models (LLMs) raises concerns regarding HIPAA compliance and data leaks. Furthermore, **Algorithmic Bias** remains a critical ethical concern; if training data lacks diversity, the AI may provide less accurate recommendations for specific demographic groups. Lastly, the **"Black Box" nature** of deep learning models makes it difficult for doctors to understand the "why" behind a diagnosis, leading to a trust gap between human clinicians and autonomous systems.

IV. FUTURE SCOPE

The future of healthcare AI lies in **Personalized Genomic Medicine**, where AI will design treatment protocols based on an individual's specific DNA sequence. We also anticipate the rise of **Ambient Intelligence**—hospital rooms equipped with non-invasive sensors that monitor patient vitals and movements without requiring wearable devices. Additionally, as **Edge AI** evolves, wearable devices (like smartwatches) will perform complex health diagnostics locally, providing real-time alerts for chronic condition management without needing a constant cloud connection.

V. CONCLUSION

AI is not a replacement for medical professionals but a powerful augmentation tool. By automating routine tasks and providing deep analytical insights, AI allows healthcare providers to focus more on patient-centric care. To achieve full integration, the industry must focus on developing **Explainable AI (XAI)** to build clinician trust and establish robust international frameworks for data ethics. It is recommended that healthcare institutions invest in "Human-in-the-Loop" systems where AI provides the insights, but the final clinical decision remains with the human expert.

3. TOOLS & SOURCE CODE

Technological Stack:

- **Framework:** LangChain (ReAct Agent)
- **LLM:** Groq (LLaMA-3.3-70B)
- **Tools:** DuckDuckGo Search, Wikipedia API